

MACHINE LEARNING PROJECT REPORT

Customer Churn

An End-to-End Predictive Analytics Pipeline for Proactive Customer Retention

Executive Summary

In the telecommunications industry, retaining an existing customer is substantially less costly than acquiring a new one. This project builds an end-to-end machine learning pipeline to predict Customer Churn — identifying subscribers likely to cancel their service — so the business can intervene with targeted retention offers before the customer leaves.

The final deployed model is a standard, untuned Logistic Regression classifier, selected after multiple rounds of iterative experimentation. It achieves an ROC-AUC of 0.8476 and correctly identifies 80% of customers who actually churn, providing the business with a reliable early-warning system for revenue-at-risk accounts.

0.8476

ROC-AUC SCORE

74%

OVERALL ACCURACY

80%

CHURN RECALL

1 Problem Identification

Customer churn — a subscriber cancelling their service — directly erodes recurring revenue and forces the business to spend disproportionately on acquisition to replace lost accounts. The objective of this project is to build a robust, interpretable machine learning pipeline that flags high-risk customers before they churn, enabling proactive, targeted retention strategies rather than reactive win-back efforts.

The Core Problem

Analysis of the underlying account data points to a more specific pattern than churn alone: high-value customers are leaving early in their relationship with the company, driven by a gap between what they expect and what they experience, an absence of contract lock-in, and friction during product setup.

- **The Expectation-Support Gap.** Customers on premium Fiber Optic plans pay an average of \$91.50 per month, yet over 70% of them lack essential support add-ons such as TechSupport or OnlineSecurity. When setup issues or speed drops occur, these customers feel they are not getting value for a premium price.
- **Low Switching Barriers.** The majority of new customers start on flexible Month-to-Month contracts paired with manual Electronic Check payments — a combination that gives frustrated customers an easy, low-friction opportunity to leave at any time.
- **Onboarding Drop-off.** New customers frequently leave before experiencing the full value of the service: over 55% of total churn occurs within the customer's first 12 months.

2 Business Impact

The impact of this churn pattern extends well beyond the raw customer count — it is concentrated in the company's most valuable segment.

- **Revenue Bleed.** Of 7,043 total customers, 1,869 (26.54%) have churned, representing a monthly revenue loss of \$139,130.85 — equivalent to 30.5% of total monthly recurring revenue.
- **Disproportionate High-Value Loss.** Premium Fiber Optic customers account for 82.15% (\$114,300) of all lost revenue. The business is retaining its low-margin DSL and landline users while losing its most profitable segment.
- **Wasted Acquisition Spend.** Acquiring premium internet subscribers carries heavy upfront sales and installation costs. With 47.4% of new customers lost within Year 1, the company frequently fails to recoup its customer acquisition cost before the customer leaves.

3 Dataset & Feature Overview

The project uses the standard Telco Customer Churn dataset, which combines demographic details, account attributes, and subscribed services for 7,043 customers.

Target Variable

- Churn — whether the customer left within the last month (Yes / No).

Key Predictive Signals

Feature-level analysis surfaced several variables with strong, business-relevant relationships to churn:

Feature	Churn Signal	Business Interpretation
InternetService	Fiber: 41.9% / DSL: 19.0%	Highest-risk segment; high cost paired with unmet support expectations.
tenure	~47% of churn in Yr 1	Strongest predictor overall; risk falls sharply the longer a customer stays.
TechSupport	Absence → higher churn	Single biggest driver of high-tier Fiber Optic churn.
SeniorCitizen	41.7% churn rate	Higher technical friction without dedicated assistance.
Contract / Payment	Month-to-month + e-check	Lowest switching barrier; easiest combination to cancel from.
Partner / Dependents	Presence → lower churn	Shared household decisions and switching friction increase loyalty.
OnlineSecurity / Backup / DeviceProtection	Presence → lower churn	Add-ons build product stickiness and protect the user experience.
gender	~26–27% (both)	Neutral factor; no meaningful difference in churn rate.
customerID	None	Arbitrary identifier; dropped before training.

Additional account fields — MonthlyCharges, TotalCharges, PhoneService, MultipleLines, and streaming add-ons (StreamingTV, StreamingMovies) — were retained as supporting features describing billing level and service depth.

4 Data Engineering & Leakage Prevention

Preparing the raw data for modeling required careful cleaning and a strict separation between training and testing data to ensure the reported performance reflects genuine predictive ability.

- **Handling Skewed Data.** A log transformation (`np.log1p`) was applied to `TotalCharges` to correct its heavily right-skewed distribution before scaling.
- **Categorical Encoding.** Categorical attributes were converted into numerical features using one-hot encoding (`pd.get_dummies`).
- **Leakage Prevention.** The dataset was split into training (80%) and testing (20%) sets before any scaling was applied. `StandardScaler` was fit exclusively on the training data, so the test set remained entirely unseen during preprocessing — preventing information from the test set from leaking into the model.

5 Iterative Experimentation & Model Selection

Several optimization paths were tested in pursuit of the strongest possible model. Each is summarized below, along with the outcome that informed the next step.

#	Approach	Outcome
1	Custom threshold tuning loop	Unstable; over-corrected false positives.
2	GridSearchCV on Logistic Regression	Tuning regularization and solvers produced a performance plateau with no meaningful metric gains.
3	Random Forest ensemble	Comparable performance to the baseline while adding unnecessary complexity.
4	Standard, untuned Logistic Regression	Final architecture. A clean, stable, and interpretable baseline that avoids the instability introduced by aggressive tuning.

This progression illustrates a disciplined, evidence-driven approach: added complexity was only justified when it produced a genuine improvement, and none of the tuning or ensemble methods outperformed the simpler baseline on this dataset.

6 Final Model Performance

The standard, untuned Logistic Regression model delivered reliable class separation and strong overall results on the held-out test set.

Classification Report

Class	Precision	Recall	F1-Score
Class 0 — Retained Customers	0.91	0.72	0.80
Class 1 — Churned Customers	0.51	0.80	0.62

Confusion Matrix

	Predicted: Retained (0)	Predicted: Churned (1)
True: Retained (0)	True Negatives	False Positives
True: Churned (1)	False Negatives	True Positives

- **ROC-AUC Score:** 0.8476 — strong separability between the churned and retained classes.
- **Overall Accuracy:** 74% across both classes.

7 Business Solution & Trade-off Analysis

The core business value of this model lies in its recall on the churn class: by correctly identifying 80% of customers who actually churn, the pipeline catches the large majority of at-risk subscribers, giving the retention team a wide net to work with.

- **The Precision Trade-off.** Precision on the churn class sits at 0.51, meaning roughly half of the customers flagged as high-risk will not actually churn. In a retention context this is an acceptable and even favorable trade-off: the cost of a missed churner (lost recurring revenue, wasted acquisition spend) is materially higher than the cost of a false alarm (a retention offer sent to a customer who was going to stay anyway).
- **Deployment Recommendation.** This configuration is well suited to real-world deployment in a proactive retention framework, particularly when paired with the segment-level findings above — prioritizing outreach toward new Fiber Optic customers without TechSupport, and month-to-month, electronic-check payers within their first 12 months.

8 Conclusion

This project demonstrates that a carefully engineered, leakage-free pipeline built on a simple, interpretable model can outperform more complex alternatives on this dataset. The standard Logistic Regression classifier was selected not because it was the most sophisticated option tested, but because it delivered the most stable and business-relevant results after Random Forest and grid-search tuning both failed to improve on it.

Combined with the feature-level insights on tenure, contract type, internet service tier, and support add-ons, the model gives the business both a working early-warning system and a clear, actionable narrative for where retention investment will have the greatest impact.